CyberShield

Cybersecurity Awareness Website

Project Report

Technology: Python + Streamlit

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1. Project Overview

Field	Details
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Project Name	CyberShield - Cybersecurity Awareness Website
Technology	Python 3.7+, Streamlit
Type	Single-Page Web Application
Difficulty Level	Beginner
Total Files	1 (app.py)
Lines of Code	~313
External Library	streamlit
Built-in Library	re (regex)

2. Objective

The goal of this project is to build a simple, interactive, single-page website that educates users ? especially beginners ? about cybersecurity fundamentals. The website provides awareness about common cyber threats, practical safety tips, a live password strength checker, and an interactive quiz ? all within a single Python file using the Streamlit framework.

3. Problem Statement

Cybercrime is growing rapidly worldwide. Many people ? especially non-technical users ? remain unaware of the basic threats they face online and how to protect themselves. There is a strong need for beginner-friendly educational tools that make cybersecurity concepts easy to understand without requiring any technical background.

CyberShield addresses this problem by delivering cybersecurity awareness content in a clean, interactive, and accessible web application.

4. Tech Stack

Layer	Technology	Purpose
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Language	Python 3.7+	Core programming language
Web Framework	Streamlit	Build and serve the web UI
Regex Engine	re (built-in)	Password pattern matching
Styling	Inline CSS	Custom dark theme and layout
State Mgmt	st.session_state	Persist quiz answers across reruns

5. Project Structure

The entire application is kept in a single file for beginner friendliness:

- ? app.py ? Complete website source code (single file)
- ? README.md ? Setup and usage guide
- ? CyberShield_Project_Report.pdf ? This report

6. Features & Modules

6.1 Homepage Header

Displays the project name CyberShield with a shield icon and subtitle. Sets the browser tab title and favicon via st.set_page_config.

6.2 Introduction to Cybersecurity

A styled card explaining what cybersecurity is in plain language, emphasising that everyone ? not just IT professionals ? should understand basic cyber hygiene.

6.3 Common Cyber Threats

Six threats are displayed as styled badge cards with a blue left-border accent:

Phishing	Fake emails/sites trick users into revealing credentials
Malware	Malicious software steals data or damages devices
Hacking	Unauthorised access to systems or networks
Online Fraud	Scammers pose as banks or officials to steal money
Weak Passwords	Short passwords are cracked instantly by automated tools
Public Wi-Fi	Unsecured networks allow data interception

6.4 Basic Safety Tips

Eight actionable tips shown as green left-bordered cards:

- ? Use passwords with 12+ characters (mix uppercase, lowercase, digits, symbols)
- ? Never reuse passwords across different websites
- ? Enable Two-Factor Authentication (2FA) wherever possible
- ? Check for https:// before entering personal data on any website
- ? Keep OS, browser, and apps updated to fix security vulnerabilities
- ? Avoid clicking suspicious links or downloading unexpected attachments
- ? Lock devices when not in use and set strong PINs
- ? Back up important data regularly to an external drive or cloud

6.5 Password Strength Checker

The check_password_strength() function evaluates passwords against 6 criteria using Python's built-in re (regex) module:

Criteria	Method	Points
At least 8 chars	len(pwd) >= 8	+1
At least 12 chars	len(pwd) >= 12	+1
Uppercase letter	re.search([A-Z], pwd)	+1
Lowercase letter	re.search([a-z], pwd)	+1
Digit	re.search(\d, pwd)	+1
Special character	re.search([!@#$...], pwd)	+1

Score	Strength	Bar Colour
0 - 2	Weak	#ef4444 (Red)
3 - 4	Medium	#f59e0b (Amber)
5 - 6	Strong	#22c55e (Green)

A visual progress bar renders the score as a percentage. Missing criteria are listed below

the bar. The password field uses type=password ? nothing is stored.

6.6 Cybersecurity Quiz

An interactive 5-question multiple-choice quiz covering key cybersecurity concepts. Uses st.form so all answers are collected before evaluation, and st.session_state to persist selections across Streamlit reruns.

Q#	Topic	Correct Answer
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Q1	Definition of phishing	A fake attempt to steal info
Q2	Strongest password example	T!9kR#2mLq@7
Q3	What 2FA stands for	Two-Factor Authentication
Q4	Secure URL indicator	https://
Q5	Protection against malware	Keep software & OS updated

Score	Result Message
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100%	Perfect score! You're a cybersecurity pro!
60-99%	Good job! Review the topics you missed.
0-59%	Keep learning ? cybersecurity is worth mastering!

7. UI / Design Decisions

Decision	Rationale
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Dark theme #0d1117	Reduces eye strain; modern GitHub-like look
Blue headings #58a6ff	High contrast, readable on dark background
Blue threat cards	Visually groups threats consistently
Green tip cards	Green conveys safety and positive action
Single-column layout	Mobile-friendly; keeps focus on content
Inline CSS only	No external files; beginner single-file approach

8. How to Run

Prerequisites

- ? Python 3.7 or higher
- ? pip (Python package manager)

Step 1 ? Install Streamlit

python -m pip install streamlit

Step 2 ? Run the application

streamlit run app.py

Step 3 ? Open in browser

http://localhost:8501

9. Limitations & Future Improvements

Current Limitations

- ? No persistence ? quiz scores/passwords not saved between sessions
- ? Static content ? threats and tips are hardcoded in app.py
- ? No user accounts ? no login, personalisation, or history tracking
- ? Only 5 quiz questions ? limited assessment scope

Possible Future Improvements

- ? Score History ? save quiz results to CSV or SQLite database

- ? Larger Question Bank ? random question selection each attempt
- ? URL Safety Checker ? detect suspicious URLs via pattern matching
- ? Light / Dark Theme Toggle
- ? Multi-language Support for non-English users
- ? Phishing Email Simulator with real-looking examples

10. Learning Outcomes

By building and exploring this project, you will learn:

- ? How to build a web application using Python and Streamlit
- ? How to use regex (re module) for pattern matching
- ? How to manage UI state with st.session_state
- ? How to apply custom CSS inside a Streamlit app
- ? Core concepts of cybersecurity awareness
- ? Best practices for password security
- ? How to structure a beginner-friendly Python project

11. Conclusion

CyberShield is a lightweight, fully functional cybersecurity awareness web application built entirely in Python using Streamlit. It demonstrates that even a single-file Python application can deliver meaningful educational content in an engaging, interactive format.

The project is ideal for beginners learning Python web development and for anyone looking to raise cybersecurity awareness in their community or organisation.

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Shield CyberShield | Stay safe. Stay informed.